Unit: Unit 0: Foundations of Algebra
Topic: Place Value

Name: ________________

Date: ____________

Homework (Jalapeno)

Q1. In ancient Babylon, a merchant had 456 gold coins. What is the value of the digit 4?
(A) 400
(B) 40
(C) 4
(D) 45
(E) 46

Q2. The great pyramid of Giza is approximately 146.5 meters tall. Rounding this number to the nearest ten, we get
(A) 150
(B) 140
(C) 146
(D) 147
(E) 140.5

Q3. A Mayan calendar has 234 days in a cycle. What is the value of the digit 3?
(A) 300
(B) 30
(C) 3
(D) 23
(E) 234

Q4. The Rosetta Stone is 0.45 meters wide. Rounding this number to the nearest hundredth, we get
(A) 0.5
(B) 0.4
(C) 0.45
(D) 0.44
(E) 0.46

Q5. A pharaoh has 1234 slaves. The number 1234 can be written in expanded form as
(A) $1 \cdot 10^3 + 2 \cdot 10^2 + 3 \cdot 10^1 + 4 \cdot 10^0$
(B) $1 \cdot 10^2 + 2 \cdot 10^1 + 3 \cdot 10^0$
(C) $1 \cdot 10^1 + 2 \cdot 10^0$
(D) $1 \cdot 10^0 + 2 \cdot 10^0$
(E) $1 + 2 + 3 + 4$

Q6. A Greek temple has 567 columns. What is the value of the digit 6?
(A) 600
(B) 60
(C) 6
(D) 56
(E) 57

Q7. The Nile River is approximately 6695 kilometers long. Rounding this number to the nearest thousand, we get
(A) 7000
(B) 6000
(C) 6695
(D) 6694
(E) 6696

Q8. An ancient Roman aqueduct is 2345 meters long. The number 2345 can be written in standard form as
(A) $2 \cdot 10^3 + 3 \cdot 10^2 + 4 \cdot 10^1 + 5 \cdot 10^0$
(B) $2 \cdot 10^2 + 3 \cdot 10^1 + 4 \cdot 10^0$
(C) $2 \cdot 10^1 + 3 \cdot 10^0$
(D) $2 + 3 + 4 + 5$
(E) 2345

Open Ended Questions (FRQ)

Q9. Write the number 945 in expanded form and explain the value of each digit.

Q10. Round the number 4721 to the nearest hundred and explain why this is the correct answer.

Chef's Tips

- Tip 1: Emphasize the importance of understanding place value in multi-digit numbers.
- Tip 2: Encourage students to use real-world examples and historical contexts to illustrate place value concepts.
- Tip 3: Provide feedback that guides students to identify and correct common mistakes in place value calculations.